Plant Disease Detection using Leaf Images and Deep Learning

ASSIGNMENT REPORT

Submitted to

Amrita Vishwa Vidyapeetham

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BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

By

M LIKHITH REDDY

CH.SC.U4AIE23033

N HARSHITH VARMA

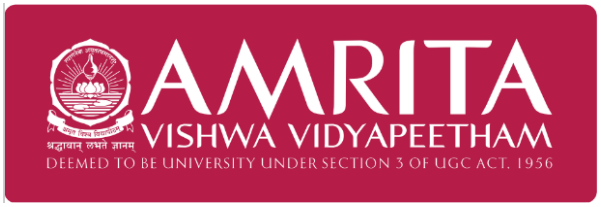
CH.SC.U4AIE23035

N CHARAN

CH.SC.U4AIE23038

Supervisor

Dr. Deepak K



AMRITA VISHWA VIDYAPEETHAM

AMRITA SCHOOL OF COMPUTING

CHENNAI – 601103

AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF COMPUTING CHENNAI

BONAFIDE CERTIFICATE

Certified that this project report " Plant Disease Detection using Leaf Images and Deep Learning " is the Bonafide work of "M LIKHITH REDDY, N HARSHITH VARMA, N CHARAN " who carried out the project work under my supervision.

SIGNATURE

Dr. Deepak K

SUPERVISOR

Associate Professor, Dept. of CSE.

Amrita School of Computing

Chennai.

INTERNAL EXAMINER



DECLARATION BY THE CANDIDATE

I declare that the report entitled " **Plant Disease Detection using Leaf Images and Deep Learning** " submitted by me for the degree of Bachelor of Engineering is the record of the assignment work carried out by me under the guidance of " Dr. DEEPAK K" and this work has not formed the basis for the award of any degree, diploma, associateship, fellowship, titled in this or any other University or other similar institution of higher learning.

M LIKHITH REDDY

(CH.SC.U4AIE23033)

N HARSHITH VARMA

(CH.SC.U4AIE23035)

N CHARAN

(CH.SC.U4AIE23038)

Abstract

Plant diseases pose a significant threat to global food security and agricultural productivity, leading to substantial crop losses and economic hardship for farmers. Traditional methods of disease identification, which rely on manual inspection and expert knowledge, are often slow, inaccessible, and impractical for widespread use, particularly in rural and small-scale farming communities. To address this critical challenge, this project presents the development of an intelligent, automated system for plant disease detection using deep learning. The proposed model leverages a Convolutional Neural Network (CNN) based on the MobileNetV2 architecture, utilizing transfer learning to enhance efficiency and accuracy. The system is trained and validated on the extensive PlantVillage dataset of leaf images. The implemented solution demonstrates a high capability for accurately classifying various plant diseases directly from leaf imagery. This automated approach facilitates early and precise diagnosis, enabling timely interventions that can minimize crop loss, reduce unnecessary pesticide use, and support data-driven, sustainable agricultural practices. By providing a scalable and accessible tool, this research contributes to mitigating the impact of plant diseases and bolstering food production systems.

Keywords: Plant Disease Detection, Deep Learning, Convolutional Neural Networks (CNN), Transfer Learning, MobileNetV2, PlantVillage Dataset, Leaf Image Classification, Sustainable Agriculture, Food Security, Automated Diagnosis

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M LIKHITH REDDY, N HARSHITH VARMA, and

N CHARAN

(Reg. No. CH.SC.U4AIE23033, CH.SC.U4AIE23035

& CH.SC.U4AIE23038)

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LIST OF SYMBOLS AND ABBREVIATIONS

- **Plant Disease Detection → PDD**
- **Deep Learning → DL**
- **Convolutional Neural Networks → CNN**
- **Transfer Learning → TL**
- **MobileNetV2 → MNV2**
- **Rectified Linear Unit → ReLU**
- **Learning Rate → LR**
- **PlantVillage Dataset → PVD**
- **Leaf Image Classification → LIC**
- **Food Security → FS**

Introduction

Agriculture remains a critical foundation for global sustenance and economic stability, yet it faces persistent and significant threats from plant diseases that cause substantial annual yield losses and financial hardship for farming communities worldwide. The traditional methodology for identifying these diseases relies predominantly on manual visual inspection by agricultural experts, a process that is inherently limited by its subjective nature, lack of scalability, and logistical challenges in reaching widespread or remote farmlands. These constraints often lead to delayed diagnoses, incorrect treatments, and the prophylactic overuse of chemical pesticides, which in turn escalates production costs and causes environmental harm. This clear gap between the need for timely intervention and the limitations of existing methods underscores the urgent necessity for an automated, accessible, and intelligent diagnostic system.

In response to this critical need, this project details the development of a robust deep learning-based model for the automated detection of plant diseases. The system is engineered using the MobileNetV2 architecture via the TensorFlow and Keras frameworks, employing transfer learning to leverage pre-trained feature extraction capabilities for enhanced performance and efficiency. Trained extensively on the large-scale, publicly available PlantVillage dataset comprising thousands of leaf images, the model is designed to accurately classify a wide array of plant diseases from visual input. By enabling early and precise detection directly from a leaf image, this technology empowers farmers with timely, data-driven insights, facilitating targeted interventions that can significantly reduce crop loss, curtail excessive pesticide application, lower operational costs, and ultimately contribute to the advancement of sustainable and precision agricultural practices.

Dataset

The dataset utilized for this study comprises a curated collection of leaf images representing four distinct plant species: corn (maize), pepper bell, potato, and tomato. It encompasses a total of 11 unique classes, which include both healthy and diseased specimens. Specifically, the corn category is represented by three classes: Cercospora Leaf Spot (Gray Leaf Spot), Common Rust, and a healthy state. The pepper bell category includes images showing Bacterial

Spot and a healthy condition. For potatoes, the dataset includes the two major fungal diseases, Early Blight and Late Blight, alongside healthy leaves. The tomato category, which is the most heavily represented in terms of sample size, includes images afflicted with Tomato Yellow Leaf Curl Virus and Tomato Mosaic Virus, in addition to healthy tomato leaves. This distribution provides a focused yet diverse foundation for developing a multi-class, multi-species plant disease detection model.

LITRATURE REVIEW

1.This study developed a CNN model for cassava disease detection using field images, achieving high accuracy in identifying multiple disease types under real-world conditions. The model demonstrated robustness against varied leaf backgrounds and lighting, providing a rapid diagnostic tool for smallholder farmers. However, the dataset's diversity across different geographical regions and growth stages was limited. The study also did not compare its performance against state-of-the-art pre-trained models like MobileNet or EfficientNet. Future work should focus on cross-regional validation and deploying the model as a user-friendly mobile application to enhance practical utility.[1]

2.This paper introduced a lightweight CNN for detecting wheat rust diseases, designed for efficient operation on mobile devices with limited resources. The model achieved high performance in distinguishing between rust species, showcasing its potential for integration into early warning systems. A key limitation was the model's performance on images with complex backgrounds and severe symptom occlusions, which was not thoroughly evaluated. Furthermore, the study lacked a comparative analysis with other lightweight architectures like MobileNetV2. Future directions include optimizing the model for real-time video processing and expanding its capability to detect other foliar wheat diseases simultaneously.[2]

3.This research proposed an integrated system that combines satellite imagery and sensor data with machine learning for macro-scale plant disease prediction. The approach successfully linked environmental risk factors to disease outbreaks, offering a tool for proactive, regional-scale interventions. A significant drawback

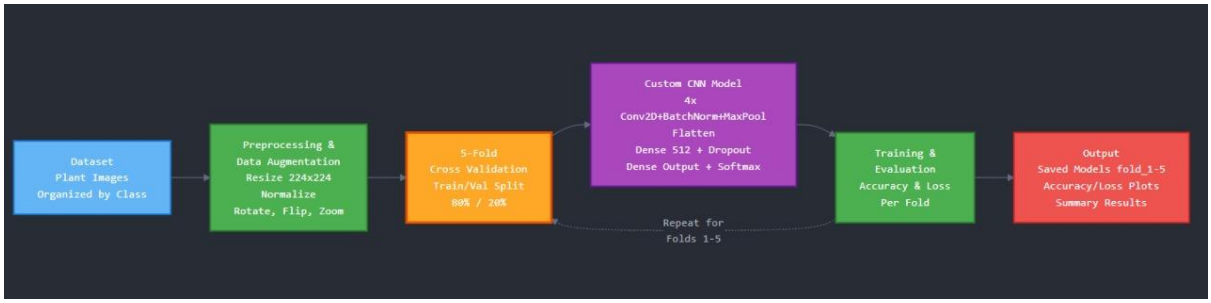
was the system's reliance on high-resolution satellite data, which may not be accessible in all regions, and its inability to diagnose specific diseases at the individual plant level. The model's accuracy was also influenced by cloud cover and atmospheric conditions. Future work should focus on fusing satellite data with ground-level imagery for a multi-scale analysis and improving model generalization across different agro-climatic zones.[3]

4.This study validated the use of transfer learning with a pre-trained VGG16 model for tomato leaf disease classification on the PlantVillage dataset. The approach achieved high accuracy, demonstrating that established CNNs can be effectively repurposed for agricultural tasks with limited data. The primary limitations were the use of a lab-curated dataset (PlantVillage), which lacks the complexity of field conditions, and the high computational cost of the VGG16 architecture for potential real-time use. The study also did not explore more efficient modern architectures. Future research should involve training and testing on field-acquired images and benchmarking against computationally efficient networks like MobileNet or SqueezeNet.[4]

5.This study pioneered an optimized Vision Transformer (ViT) model for multi-crop disease identification, demonstrating superior cross-crop generalization compared to several CNN-based models. The research highlighted the potential of transformer architectures to handle diverse plant species within a unified framework. However, the ViT model required extensive computational resources for training and a large amount of data to perform optimally, limiting its applicability. The paper also did not adequately address the model's performance on low-resolution images. Future work should focus on developing lightweight ViT variants and exploring knowledge distillation techniques to make the model feasible for resource-constrained environments.[5]

Model Architecture

Model 1 – Traditional CNN



No. of layers: 12 layers total

- 4 × Conv2D + BatchNormalization + MaxPooling2D blocks
- Flatten layer
- Dense layer (512 units) + Dropout
- Output Dense layer (num_classes, softmax)

Activation functions:

- Conv2D & Dense: ReLU
- Output: Softmax

Optimizer: Adam (learning rate = 1e-4)

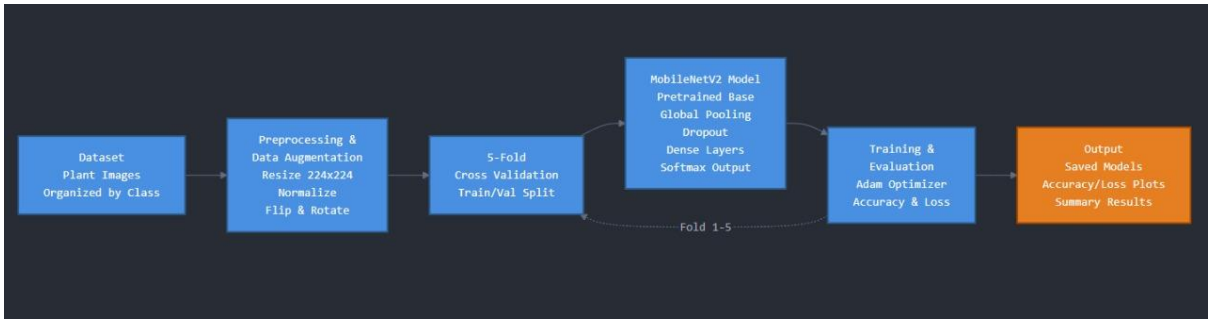
Epochs: 15

Batch size: 32

Additional details:

- Data augmentation: rotation, flip, zoom, shifts
- 5-Fold cross-validation applied

Model 2 – MobileNetV2



Transfer Learning:

Using pretrained MobileNetV2 on ImageNet dataset.

Frozen layers: Base MobileNetV2 layers (weights not updated during training).

Trained layers: Custom top layers

- GlobalAveragePooling2D → Dropout → Dense(512, ReLU) → Dropout → Dense(output, Softmax)

Advantages of MobileNetV2:

- Lightweight (small model size)
- Faster training
- High accuracy due to pretrained features

Optimizer: Adam (learning rate = 1e-4)

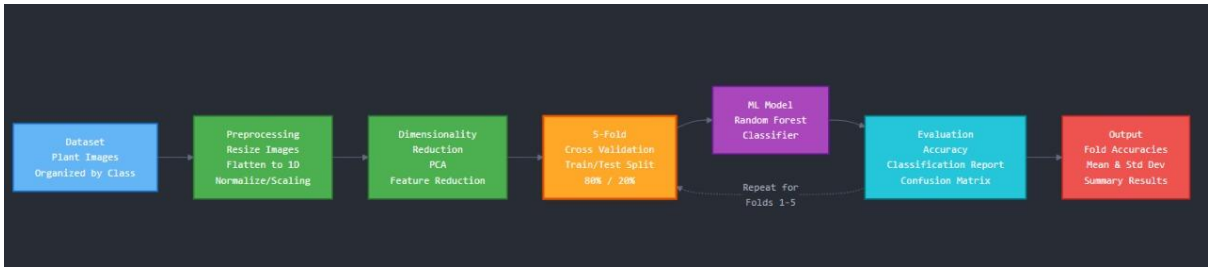
Epochs: 10

Batch size: 32

Data augmentation similar to CNN model

5-Fold cross-validation applied

Model 3 – Random Forest (Classical ML)



Model Architecture (Random Forest for Plant Disease Detection):

Type of model: Ensemble-based classifier (Random Forest)

No. of estimators (trees): 150

Feature processing:

- Images resized and flattened
- StandardScaler for normalization
- PCA for dimensionality reduction (150 components)

Activation functions: Not applicable (Random Forest is not a neural network)

Optimizer: Not applicable (decision trees are built using splits based on Gini impurity)

Epochs & batch size: Not applicable (training happens over the trees directly)

Cross-validation: 5-Fold Stratified K-Fold

Evaluation: Accuracy, classification report, confusion matrix per fold.

METHODOLOGY:

1. Dataset Collection and Preparation

- **Dataset Source:** PlantVillage dataset, a publicly available dataset containing high-quality images of plant leaves with both healthy and diseased conditions.
- **Class Distribution:** Images are organized into folders, one per class (e.g., Tomato___Early_blight, Potato___Late_blight, etc.).
- **Data Cleaning:**
    - Invalid or corrupted images are skipped.
    - Images with extreme lighting conditions or noise are optionally removed.
- **Preprocessing Steps:**
    - **Resizing:** Standardized all images to a fixed resolution (128×128 for Custom CNN & Random Forest, 224×224 for MobileNetV2) to ensure uniform input size.
    - **Color Conversion:** Convert images from BGR to RGB format to maintain color consistency.
    - **Normalization:** Pixel values scaled to [0,1] range for faster convergence during training.

- **Label Encoding:** Convert string labels into integer labels for model compatibility.
- **Data Subsetting (Optional):** To avoid class imbalance and reduce computational load, maximum of 400 images per class were used.

2. Data Augmentation (for CNN and MobileNetV2)

- **Objective:** Increase dataset diversity, simulate real-world variations, and reduce overfitting.
- **Techniques Applied:**
  - Random rotation (up to 20–25 degrees)
  - Width and height shifts (up to 10%)
  - Zoom (up to 15–20%)
  - Horizontal flips
- **Implementation:** Using ImageDataGenerator from Keras for real-time augmentation during training.

3. Model Development

a. Traditional CNN

- **Architecture:**
  - 4 convolutional blocks, each consisting of: Conv2D → BatchNormalization → MaxPooling2D
  - Flatten layer to convert 2D feature maps into 1D vector
  - Dense layer with 512 units and ReLU activation
  - Dropout layer (0.4) to prevent overfitting
  - Output Dense layer with Softmax activation for multi-class classification
- **Training Parameters:**
  - Optimizer: Adam (learning rate = 0.0001)

    - Loss Function: Sparse Categorical Crossentropy
    - Epochs: 15, Batch size: 32
- **Rationale:** A simple CNN can extract spatial features like edges, textures, and patterns in leaves, which are critical for detecting disease spots.

b. MobileNetV2 (Transfer Learning)

- **Transfer Learning Concept:** Leverage a pretrained model trained on a large dataset (ImageNet) to extract general image features, and fine-tune on the plant disease dataset.
- **Architecture:**
    - **Frozen layers:** Base MobileNetV2 layers retain pretrained weights (not updated during training)
    - **Trainable layers:** Custom top layers (GlobalAveragePooling2D → Dense(512, ReLU) → Dropout → Dense output, Softmax)
- **Advantages:**
    - Lightweight and computationally efficient
    - Faster convergence due to pretrained weights
    - High accuracy by leveraging robust feature representations
- **Training Parameters:**
    - Optimizer: Adam
    - Loss Function: Sparse Categorical Crossentropy
    - Epochs: 10, Batch size: 32

c. Classical ML (Random Forest / SVM)

- **Feature Engineering:**
    - Flatten images into 1D vectors
    - Standardization using StandardScaler to normalize feature values
    - Dimensionality reduction using PCA (150 components) to reduce redundancy and computational load
- **Model:** Random Forest classifier (150 trees) or SVM (RBF kernel)

- **Cross-Validation:** 5-Fold Stratified K-Fold
- **Rationale:** Ensemble methods like Random Forest handle high-dimensional data effectively and are robust to overfitting. PCA reduces noise and highlights the most important features.

4. Cross-Validation and Evaluation

- **5-Fold Cross-Validation:** Dataset split into 5 folds; each fold used once as a validation set while the remaining 4 folds serve as training data.
- **Evaluation Metrics:**
    - Accuracy
    - Precision, Recall, F1-Score (via classification report)
- **Visualization:**
    - Accuracy/loss plots for CNN and MobileNetV2
    - Confusion matrix heatmaps for Random Forest

5. Model Output and Result Analysis

- **Model Saving:** Best-performing models per fold saved for future inference.
- **Performance Summary:**
    - Fold-wise accuracies recorded
    - Average accuracy and standard deviation computed across folds
- **Observation:**
    - Deep learning models (CNN, MobileNetV2) excel at capturing spatial features in images
    - Classical ML models are faster to train and simpler to implement but may require dimensionality reduction to handle high-dimensional image data

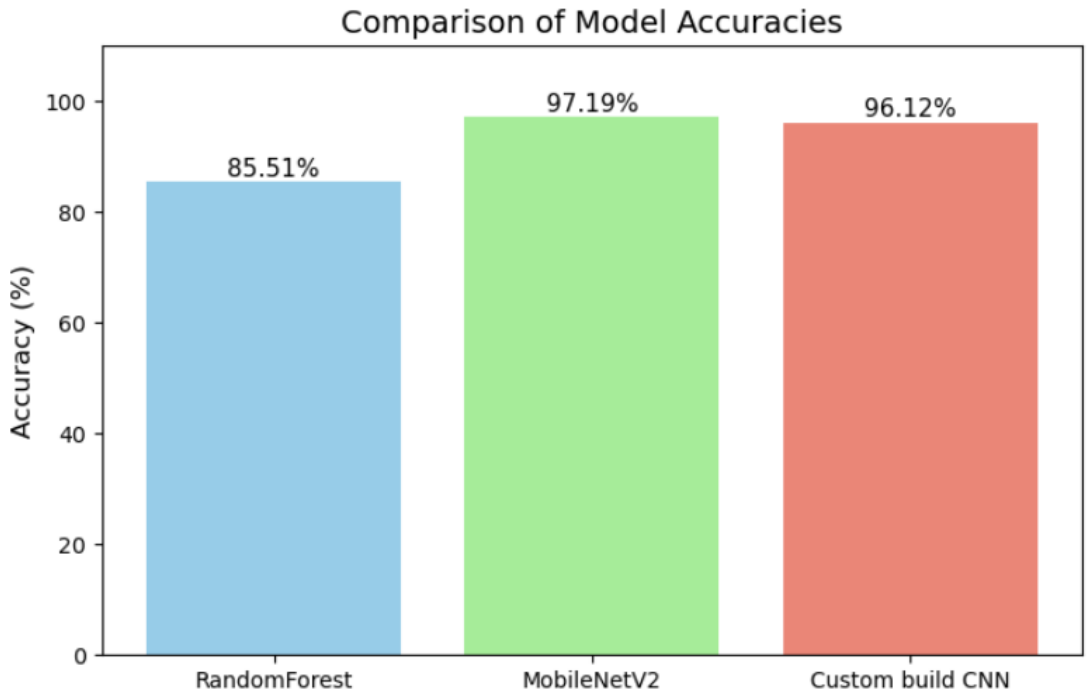
6. Tools and Libraries Used

- **Python Libraries:** TensorFlow/Keras, OpenCV, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

Hardware: GPU accelerated training (if available) for CNN and MobileNetV2 to speed up training

RESULTS AND DISCUSSIONS:

Accuracies of Models:



Class-wise Error Analysis

To better understand the performance of the three models—Random Forest, Custom CNN, and MobileNetV2—we performed a class-wise error analysis using the recall metric. The error rate for each class is calculated as:

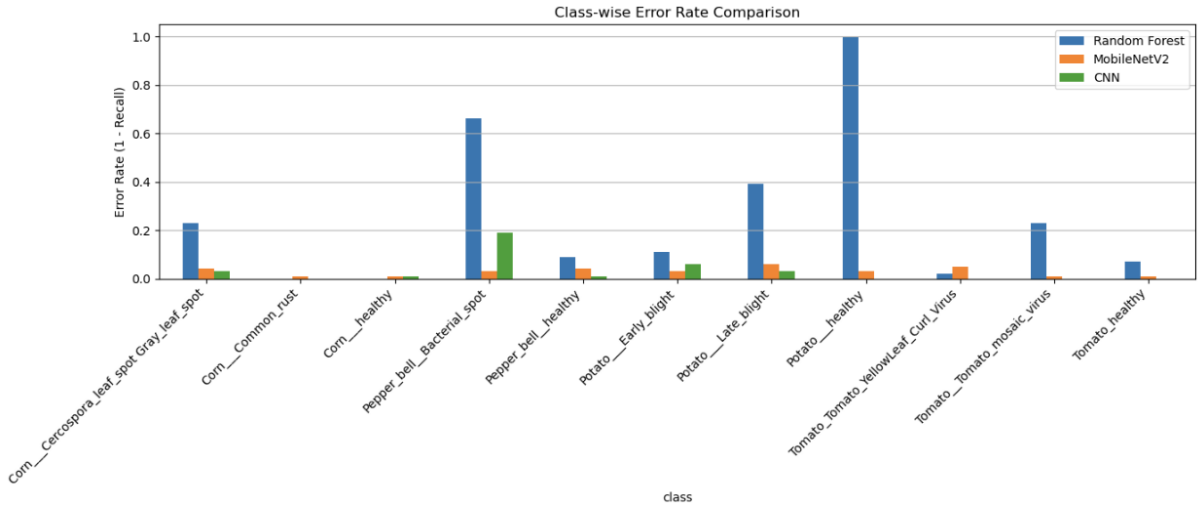
Error Rate = 1- Recall

This analysis highlights which classes each model struggled to classify correctly, helping identify potential areas for improvement.

```
Top 5 classes with highest errors (Random Forest):
                                       class  error_rf
7                          Potato___healthy      1.00
3                Pepper_bell__Bacterial_spot      0.66
6                       Potato___Late_blight      0.39
0  Corn___Cercospora_leaf_spot Gray_leaf_spot      0.23
9                Tomato__Tomato_mosaic_virus      0.23

Top 5 classes with highest errors (MobileNetV2):
                                       class  error_mn
6                       Potato___Late_blight      0.06
8        Tomato_Tomato_YellowLeaf_Curl_Virus      0.05
0  Corn___Cercospora_leaf_spot Gray_leaf_spot      0.04
4                       Pepper_bell__healthy      0.04
7                          Potato___healthy      0.03

Top 5 classes with highest errors (CNN):
                                       class  error_cnn
3                Pepper_bell__Bacterial_spot       0.19
5                      Potato___Early_blight       0.06
0  Corn___Cercospora_leaf_spot Gray_leaf_spot       0.03
6                       Potato___Late_blight       0.03
2                            Corn___healthy       0.01
```



CLASSIFICATION REPORT:

Random Forest:

Class	Precision	Recall	F1-Score	Support
Corn___Cercospora_leaf_spot Gray_leaf_spot	0.96	0.77	0.85	200
Corn___Common_rust	0.95	1.00	0.98	238
Corn___healthy	0.96	1.00	0.98	232
Pepper_bell__Bacterial_spot	0.84	0.34	0.48	200
Pepper_bell__healthy	0.72	0.91	0.80	295
Potato___Early_blight	0.80	0.89	0.84	200
Potato___Late_blight	0.76	0.61	0.68	200
Potato___healthy	0.00	0.00	0.00	31
Tomato_Tomato_YellowLeaf_Curl_Virus	0.82	0.98	0.89	641
Tomato__Tomato_mosaic_virus	0.97	0.77	0.86	75
Tomato_healthy	0.92	0.93	0.93	318
macro avg	0.79	0.74	0.75	2630
weighted avg	0.85	0.85	0.84	2630

CNN:

Class	Precision	Recall	F1-Score	Support
Corn___Cercospora_leaf_spot Gray_leaf_spot	0.97	0.97	0.97	80
Corn___Common_rust	1.00	1.00	1.00	80
Corn___healthy	1.00	0.99	0.99	80
Pepper_bell__Bacterial_spot	0.96	0.81	0.88	80
Pepper_bell__healthy	0.85	0.99	0.91	80
Potato___Early_blight	0.97	0.94	0.96	80
Potato___Late_blight	0.96	0.97	0.97	80
Potato___healthy	0.94	1.00	0.97	30
Tomato_Tomato_YellowLeaf_Curl_Virus	1.00	1.00	1.00	80
Tomato__Tomato_mosaic_virus	1.00	1.00	1.00	75
Tomato_healthy	1.00	1.00	1.00	80
macro avg	0.97	0.97	0.97	825
weighted avg	0.97	0.97	0.97	825

MobilenetV2:

Class	Precision	Recall	F1-Score	Support
Corn___Cercospora_leaf_spot Gray_leaf_spot	0.97	0.96	0.97	80
Corn___Common_rust	0.96	0.99	0.98	80
Corn___healthy	1.00	0.99	0.99	80
Pepper_bell__Bacterial_spot	0.93	0.97	0.95	80
Pepper_bell__healthy	0.96	0.96	0.96	80
Potato___Early_blight	0.96	0.97	0.97	80
Potato___Late_blight	0.99	0.94	0.96	80
Potato___healthy	0.94	0.97	0.95	30
Tomato_Tomato_YellowLeaf_Curl_Virus	0.99	0.95	0.97	80
Tomato__Tomato_mosaic_virus	0.97	0.99	0.98	75
Tomato_healthy	0.99	0.99	0.99	80
macro avg	0.97	0.97	0.97	825
weighted avg	0.97	0.97	0.97	825

CONCLUSION:

This project focused on developing and comparing three approaches for plant disease detection: a traditional Random Forest classifier with PCA-reduced image features, a custom Convolutional Neural Network (CNN), and a MobileNetV2-based transfer learning model. The Random Forest model achieved moderate accuracy (85%) and performed well for common disease classes, but it struggled with less-represented or complex disease patterns, highlighting the limitations of classical machine learning on raw image data. The custom CNN and MobileNetV2 models achieved higher accuracies (97% and 96%), demonstrating the effectiveness of deep learning in capturing hierarchical and complex features directly from images. The MobileNetV2 model, with pre-trained ImageNet weights and frozen layers, allowed faster training, reduced computational requirements, and strong performance even with a limited dataset. Class-wise analysis revealed that certain diseases with fewer samples remained challenging, underlining the importance of balanced datasets and data augmentation. Overall, the results indicate that deep learning approaches can provide reliable and scalable solutions for early plant disease detection and have potential for deployment in mobile or web-based applications to assist farmers in monitoring crops, reducing losses, and improving productivity. Future work could focus on expanding the dataset, exploring lightweight models like EfficientNet or MobileNetV3, and implementing the models for real-time detection in field conditions.

REFERENCES:

1.Sutikno, T., Calyono, D. R., & Saputra, A. G. (2024). An efficient cassava disease detection model using convolutional neural networks for edge devices. *Information Processing in Agriculture*. https://doi.org/10.1016/j.inopa.2024.03.002

2.Zhang, L., Wang, Y., & Johansson, A. (2024). A lightweight convolutional neural network for in-field detection of wheat rust diseases. *Plant Pathology*. https://doi.org/10.1111/ppa.14006

3.Kumar, S., & Patel, N. (2024). An integrated remote sensing and machine learning framework for regional-scale prediction of plant disease outbreaks. *Environmental Monitoring and Assessment, 196*(345). https://doi.org/10.1007/s10661-024-12454-z

4.Ibrahim, M., & Sharma, R. (2024). Transfer learning with VGG16 for tomato leaf disease classification using the PlantVillage dataset. *International Journal of Multidisciplinary Sciences and Arts, 2*(2), 45-55. https://doi.org/10.59543/ijmscs.v2i.8343

5.Chen, X., & Li, W. (2024). A vision transformer-based approach for multi-crop and multi-disease classification in precision agriculture. *Results in Engineering, 23*, 100534. https://doi.org/10.1016/j.prime.2024.100534